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Clinical, demographic, and lifestyle characteristics of patients with cervicogenic headache: comparison with an age- and sex-matched group of individuals with migraine

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Abstract

Background Cervicogenic headache (CEH) can be frequently misdiagnosed as migraine. This study aimed to compare clinical, demographic, and lifestyle characteristics of patients with CEH and individuals with migraine.

Methods The retrospective cross-sectional study included 112 patients with CEH and 112 age- and sex-matched individuals with migraine consulted at a single headache center in 2022–2026. The analysis involved clinical and demographic characteristics, and lifestyle factors, among them dominant sleep posture.

Results Compared with individuals with migraine, patients with CEH were diagnosed with headache at an older age (36.27 ± 11.95 vs. 22.62 ± 11.13 years, $p < 0.001$), had shorter duration of disease (4.08 ± 5.40 vs. 17.74 ± 11.60 years, $p < 0.001$), more often reported neck pain, also occurring independently of headache (76.92% vs. 39.56%, $p < 0.001$), more frequently presented with greater occipital nerve (GON) sensitivity to palpation (85.44% vs. 30.21%, $p < 0.001$) and vitamin B12 deficiency (56.86% vs. 9.38%, $p < 0.001$). The CEH group contained a larger proportion of active smokers (43.75% vs. 10.71%, $p < 0.001$) and belly (prone) sleepers (40.18% vs. 14.29%, $p < 0.001$) than the migraine group. A multivariate regression analysis identified belly sleeping (OR = 11.10, $p = 0.006$), smoking (OR = 22.61, $p = 0.005$), and vitamin B12 deficiency (OR = 8.58, $p = 0.005$) as independent determinants of CEH.

Conclusions CEH differs from migraine in terms of selected clinical, demographic, and lifestyle characteristics. Neck pain independent of headache, GON sensitivity to palpation, older age at the onset of headache, and shorter time elapsed from the first manifestation to referral might be diagnostic prompts in suspected CEH. Belly sleeping, vitamin B12 deficiency, and smoking appear as independent determinants of CEH and might be considered in the prevention and management of this condition.

Keywords Migraine disorders, Cervicogenic headache, Cross-sectional study, Sleep posture, Trigemino-cervical nucleus, Neck pain, Vitamin B12 deficiency, Smoking, Differential diagnosis, Prevention

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Background

Cervicogenic headache (CEH) is a secondary headache disorder originating from the neck structures. The key structure involved in the pathogenesis of CEH is the trigeminocervical nucleus, where cervical sensory nerves transmitting pain signals from affected neck structures converge with trigeminal fibers [1, 2]. As a result of the convergence of neural signals, pain originating in the neck may also involve cranial regions, typically innervated by the trigeminal system [3, 4].

The prevalence of CEH remains unknown. According to various sources, CEH may constitute from 0.4% to 42% of all headache cases among individuals over 50 years of age [5, 6]. Such a wide spectrum of prevalence may result from the lack of unified diagnostic criteria for CEH. Aside from the International Headache Society criteria for CEH applied by headache specialists worldwide [7], several other diagnostic criteria have been published, among them those by the Cervicogenic Headache International Study Group [8, 9], which may create confusion among healthcare professionals.

Additionally, CEH is diagnosed and managed not only by headache experts, but also by general neurologists, neurosurgeons, orthopedic surgeons and general surgeons, anesthesiologists, general and internal medicine specialists, psychiatrists, physiotherapists, rehabilitation and sports medicine specialists, which also contributes to discrepancies in epidemiological data and standards of care [10].

Further, CEH can be frequently misdiagnosed as migraine and vice versa, with a misdiagnosis rate up to 50% [11]. The high misdiagnosis rate may be a consequence of a convergence of sensory information in the trigeminocervical nucleus; as a result, not only can neck pain be perceived as headache, as in CEH, but headache may be accompanied by neck pain in up to 77% of individuals with migraine [12–15]. Thus, some patients may not receive optimal treatment.

Given these diagnostic challenges, detailed history taking and careful clinical evaluation are key in distinguishing between CEH and migraine [13]. Approximately 400 new cases of headache are consulted at our center annually, among them approximately 10% of CEH cases. We have developed a standardized protocol for history taking over the years, and hence, a unified dataset is available for each person with a headache treated at our center.

The aim of our study was to compare clinical, demographic, and lifestyle characteristics in patients with CEH and age- and sex-matched individuals with migraine.

Methods

Study design

A retrospective cross-sectional study included the data of patients consulted at our Headache Center in Bydgoszcz

(Central Poland) between 2022 and 2026. The study protocol was approved by the Local Ethics Committee at the Ludwik Rydygier Collegium Medicum in Bydgoszcz. No written consent was required due to the retrospective character of the study. The study was conducted in accordance with the Declaration of Helsinki. The results are reported according to the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) Statement [16].

Participants

The study included patients with CEH and age- and sex-matched individuals with migraine. Participants were eligible if they were 18 years of age or older and had a diagnosis of migraine with or without aura, or a diagnosis of CEH according to the 3rd edition of the International Classification of Headache Disorders (ICHD-3) [7]. Individuals with either episodic or chronic migraine were eligible, including those who were on prophylactic medications. Individuals with co-existing CEH and migraine, and those in whom CEH or migraine co-existed with another ICHD-3 diagnosis, including post-traumatic headache, were excluded from the analysis. Patients with medication overuse headache, as defined by ICHD-3, were not excluded. Except for fibromyalgia, severe or unstable psychiatric diseases, severe cognitive deficits, and other serious systemic illnesses, comorbidities were not an exclusion criterion from the study.

Variables

Data from the baseline visits and control visits were extracted from the electronic medical database. At the baseline visit, all participants underwent a detailed history-taking and clinical evaluations according to the standard protocol developed at our center.

The analyzed data from medical records included: headache onset age (in years), type of headache according to the ICHD-3 (migraine or CEH), location of pain (side-locked if it occurred on the same side of the head for more than 90% of attacks, unilateral variable if it was unilateral but changed side between attacks or during the attack, or bilateral otherwise), quality of pain (pulsating or non-pulsating), duration of disease (the time elapsed from the CEH/migraine onset to referral to our center, in years), mean duration of headache episode (in hours), the presence of symptoms accompanying headache (nausea, vomiting, photo- and phonophobia, cranial autonomic symptoms), the presence of neck pain (accompanying headache or occurring independently), greater occipital nerve (GON) sensitivity to palpation (sensitivity to direct pressure of approximately 1 kg/cm² applied onto the occipital area by an experienced neurologist, M.N.), monthly headache days, acute medication days (the number of days per month on which analgesics were taken),

the presence of medication overuse headache, headache intensity measured by numeric rating scale (NRS), family history of headache, comorbidities (depression, anxiety, arterial hypertension, sleep disorders, anemia, dizziness or vertigo, autoimmune diseases, gastrointestinal disorders, bruxism), a history of COVID, vitamin B12, D3 and folic acid levels, the presence of vitamin B12 and vitamin D3 deficiency (blood levels below 180 pg/ml and 30 ng/ml, respectively), and the presence of brain (MRI) abnormalities (hyperintensity, cyst(s), atrophy, vascular malformations, or arterial hypoplasia/aplasia).

Aside from the clinical parameters mentioned above, patients' lifestyle characteristics were analyzed, including smoking, caffeine intake, shift work, and dominant sleep position.

A patient was considered a smoker if he/she reported having at least one cigarette daily within the last six months. A patient was considered a coffee drinker if he/she reported having at least one cup of caffeinated coffee (or one serving of another caffeine-containing drink) daily within the last six months. Additionally, the number of coffee cups consumed on a usual day was recorded as a continuous variable.

Each person was asked about their dominant sleep position over the last five years. Participants were considered side sleepers if they slept primarily on their left or right side; as back (supine) sleepers if they slept predominantly on their back with their spine, neck, and head in a relatively neutral alignment; and as belly (prone) sleepers if they slept mainly lying face down with their chest and abdomen against the mattress and their head twisted to either side. Individuals who declared sleeping in more than one position mentioned above were classified as variable sleepers.

Statistics

The sample size was determined based on the number of patients with CEH available for the analysis, i.e., satisfying the inclusion and exclusion criteria of the study. The study was designed as a cross-sectional retrospective analysis, and no formal sample size estimation was carried out beforehand. Continuous variables were presented using arithmetic means, standard deviations (SD), and 95% confidence intervals (95%CI), while categorical variables were presented as numbers and percentages. Missing variables were classified as missing completely at random and treated by listwise deletion during the statistical analysis. Normal distribution of continuous variables was verified with the Shapiro-Wilk test. Depending on the type of distribution, between-group comparisons of continuous variables were conducted with the Student's t-test or Mann-Whitney U-test. The distributions of categorical variables were compared using the chi-square or Fisher's exact test, whichever was more suitable

for the observation number. All variables with $p \leq 0.10$ in between-group comparisons were included in univariate logistic regression analyses as potential determinants of CEH. The results were presented as odds ratios (OR) along with their 95% CIs. Selected variables were included in a multivariate logistic regression model to verify whether they were independent determinants of CEH. Explanatory variables included in the model were tested for multicollinearity using the uncentered variance inflation factors (VIFs), and the model fit was estimated based on McFadden's pseudo- R^2 value. All statistical calculations were carried out with Stata 18.0 SE package (Stata-Corp LLC, College Station, TX, USA), with the threshold of statistical significance for all tests set at $p \leq 0.05$.

Results

The study included 112 patients with CEH and 112 age- and sex-matched individuals with migraine. Each group consisted of 89 women (79.46%) and 23 men (20.54%), with a mean age of 40.36 ± 11.52 years (range 18–65 years).

All 112 patients from the CEH group had clinical ($n=107$, 95.54%) and/or imaging evidence ($n=22$, 19.64%) of a disorder or lesion within the cervical spine or soft tissues of the neck, known to be able to cause headache. In 90 out of 112 CEH patients (80.36%), the presence of a disorder/lesion within the cervical spine or soft tissues of the neck was demonstrated solely based on clinical presentation, namely GON sensitivity to palpation ($n=76$, 67.86%) and/or limitations of passive neck movements ($n=68$, 60.71%) or resistance to passive neck movements ($n=22$, 19.64%) and/or abnormal tenderness of neck muscles ($n=14$, 12.50%) and/or uneven muscle tone between the right and left sides of the neck ($n=5$, 4.46%). In another 17 CEH patients (15.18%), the clinical evidence of a disorder/lesion within the cervical spine or soft tissues of the neck, such as limitations of passive neck movements ($n=16$, 14.29%) or resistance to passive neck movements ($n=1$, 0.89%) and/or GON sensitivity to palpation ($n=12$, 10.71%), co-existed with an imaging evidence, such as abnormal cervical lordosis ($n=14$, 12.50%) and/or uncovertebral arthrosis in C2-C3 ($n=12$, 10.71%) or disc bulging in C2-C3 ($n=1$, 0.89%). Finally, 5 CEH patients (4.46%) had solely the imaging evidence of a disorder/lesion within the cervical spine or soft tissues of the neck, namely abnormal cervical lordosis ($n=5$, 4.46%) and bilateral uncovertebral arthrosis in C2-C3 ($n=3$, 2.68%) or disc bulging in C2-C3 ($n=2$, 1.79%).

In 45 (40.18%) CEH patients, headache has developed in temporal relation to the onset of the cervical disorder or appearance of the lesion. In 33 (29.46%) patients, headache has significantly improved or resolved in parallel with improvement in or resolution of the cervical disorder or lesion. In 105 (93.75%) patients, cervical range

of motion was reduced, and headache was made significantly worse by provocative maneuvers. In 67 (59.82%) patients, headache was abolished following diagnostic blockade of a cervical structure or its nerve supply.

The results of between-group comparisons are summarized in Tables 1 and 2. CEH manifested first at a significantly older age than migraine (36.27 ± 11.95 vs. 22.62 ± 11.13 years, $p < 0.001$), and duration of disease was significantly shorter in the CEH group than in the migraine group (4.08 ± 5.40 vs. 17.74 ± 11.60 years, $p < 0.001$).

Compared with individuals with migraine, patients with CEH had a significantly shorter mean duration of headache episode (22.80 ± 26.77 vs. 54.55 ± 22.54 h,

$p < 0.001$). Individuals with CEH presented with side-locked pain significantly more often than those with migraine (87.50% vs. 71.43%, $p = 0.003$), and unlike the latter, none of them reported pulsating pain. Patients with CEH reported neck pain (associated with headache or not: 78.57% vs. 37.50%, $p < 0.001$, and 76.92% vs. 39.56%, $p < 0.001$, respectively) and GON sensitivity to palpation (85.44% vs. 30.21%, $p < 0.001$) significantly more often than individuals with migraine, and significantly less frequently presented with cranial autonomic symptoms (12.50% vs. 35.71%, $p < 0.001$). The prevalence of self-reported dizziness or vertigo was significantly lower in the individuals with CEH (1.79% vs. 8.93%, $p = 0.018$).

Table 1 Quantitative clinical and lifestyle characteristics of participants with CEH and migraine

Variable	Cervicogenic headache (n = 112)		Migraine (n = 112)		p-value	Comment*	
	n	%	n	%			
Side-locked pain	98	87.50	80	71.43	0.003		
Pulsating pain	0	0.00	89	79.46	< 0.001		
Cranial autonomic symptoms	14	12.50	40	35.71	< 0.001		
Neck pain accompanying headache	88	78.57	42	37.50	< 0.001		
Neck pain independent of headache	80	76.92	36	39.56	< 0.001	n = 104	n = 91
GON sensitivity on palpation	88	85.44	29	30.21	< 0.001	n = 103	n = 96
Medication overuse headache	3	2.68	29	26.61	< 0.001		n = 109
Family history of headache	37	33.04	85	75.89	< 0.001		
Thyroid disorders	30	26.79	23	20.54	0.271		
Depression	11	9.82	20	17.86	0.082		
Anxiety	14	12.50	15	13.39	0.842		
Disorders of sleep	44	41.90	33	36.26	0.420	n = 105	n = 91
Hormonal contraceptives	12	13.48	21	22.34	0.119	n = 89	n = 94
Arterial hypertension	11	9.91	10	8.93	0.802	n = 111	
Anemia	5	4.46	8	7.14	0.285		
Dizziness or vertigo	2	1.79	10	8.93	0.018		
Autoimmune disorders	2	1.79	4	3.57	0.408		
Intestinal disorders	15	13.39	12	10.71	0.538		
Bruxism	38	35.19	29	26.13	0.146	n = 108	n = 111
Brain MRI abnormalities	26	34.67	33	35.11	0.952	n = 75	n = 94
Vitamin B12 deficiency	29	56.86	3	9.38	< 0.001	n = 51	n = 32
Vitamin D3 deficiency	22	51.16	11	42.31	0.475	n = 43	n = 26
Folate deficiency	9	56.25	5	41.67	0.445	n = 16	n = 12
Active smoking	49	43.75	12	10.71	< 0.001		
Drinking coffee	73	65.18	87	77.68	0.038		
Shift work	10	8.93	8	7.14	0.403		
Variable sleepers	25	22.32	22	19.64	0.623		
Side sleepers	40	35.71	68	60.71	< 0.001		
Unilateral side sleepers	29	72.50	21	30.88	< 0.001	n = 40	n = 68
Belly sleepers	45	40.18	16	14.29	< 0.001		
Unilateral belly sleepers	33	73.33	9	56.25	0.205	n = 45	n = 16
Back sleepers	2	1.79	6	5.36	0.140		
Side or belly sleepers	85	75.89	84	75.00	0.877		
Unilateral side/belly sleepers	63	74.12	30	35.71	< 0.001	n = 85	n = 84

*n = 112 unless stated otherwise. Brain MRI abnormalities correspond to the presence of at least one of the following: hyperintensity, cyst(s), atrophy, vascular malformations, arterial hypoplasia/aplasia. Vitamin B12 deficiency corresponds to levels below 180 pg/ml. Vitamin D3 deficiency corresponds to levels below 30 ng/ml. GON – greater occipital nerve, MRI – magnetic resonance imaging

Table 2 Quantitative demographic, clinical, and lifestyle characteristics of participants with CEH and migraine

Variable	Cervicogenic headache (n = 112)		Migraine (n = 112)		p-value	Comment*	
	Mean (SD)	95%CI	Mean (SD)	95%CI			
Age at diagnosis (years)	36.27 (11.95)	34.03–38.51	22.62 (11.13)	20.54–24.70	<0.001		
Duration of disease (years)	4.08 (5.40)	3.07–5.09	17.74 (11.60)	15.56–19.91	<0.001		
Mean duration of headache episode (hours)	22.80 (26.77)	16.59–29.00	54.55 (22.54)	49.85–59.24	<0.001	n = 74	n = 91
Pain intensity (NRS)	7.04 (1.37)	6.78–7.29	8.54 (1.43)	8.27–8.80	<0.001		
Monthly headache days	17.07 (9.53)	15.29–18.85	13.33 (9.38)	11.58–15.09	0.003		
Acute medication days	5.45 (5.13)	4.49–6.42	11.42 (8.90)	9.76–13.09	<0.001		
Vitamin B12 (pg/ml)	237.39 (167.52)	185.84–288.95	432.54 (229.46)	333.31–531.77	<0.001	n = 43	n = 23
Vitamin D3 (ng/ml)	27.13 (12.07)	22.70–31.56	31.03 (16.01)	23.06–38.99	0.339	n = 31	n = 18
Number of COVID infections	0.84 (0.98)	0.65–1.02	0.92 (0.94)	0.76–1.08	0.503	n = 111	
Daily cups of coffee	2.20 (1.08)	1.95–2.46	1.82 (0.93)	1.63–2.01	0.015	n = 73	n = 92

*n = 112 unless stated otherwise. NRS – Numeric Rating Scale

Table 3 Demographic, clinical, and lifestyle determinants of CEH: results of univariate logistic regression analysis

Variable	OR	95%CI	p-value
Age at diagnosis (years)	1.10	1.07–1.14	<0.001
Duration of disease (years)	0.82	0.77–0.86	<0.001
Average headache episode (hours)	0.95	0.94–0.97	<0.001
Side-locked pain	2.80	1.40–5.60	0.004
Pulsating pain	(-)	(-)	(-)
Cranial autonomic symptoms	0.26	0.13–0.51	<0.001
Neck pain accompanying headache	6.11	3.38–11.04	<0.001
Neck pain independent of headache	5.09	2.74–9.47	<0.001
GON sensitivity on palpation	13.55	6.73–27.29	<0.001
Numeric Rating Scale	0.49	0.39–0.61	<0.001
Monthly headache days	1.04	1.01–1.07	0.004
Acute medication days	0.87	0.82–0.92	<0.001
Medication overuse headache	0.08	0.02–0.26	<0.001
Family history of headache	0.16	0.09–0.28	<0.001
Depression	0.50	0.23–1.10	0.086
Vertigo	0.18	0.04–0.87	0.032
Vitamin B12 deficiency	12.74	3.43–47.30	<0.001
Vitamin B12 (pg/ml)	0.99	0.99–1.00	0.003
Active smoking	6.48	3.20–13.13	<0.001
Coffee drinking	0.54	0.30–0.97	0.040
Daily cups of coffee	1.47	1.07–2.03	0.017
Side sleepers	0.36	0.21–0.62	<0.001
Unilateral side sleepers	5.90	2.49–14.00	<0.001
Belly sleepers	4.03	2.10–7.72	<0.001
Unilateral side/belly sleepers	5.15	2.67–9.97	<0.001

(-) Analysis cannot be conducted if n in at least one cell equals zero. Vitamin B12 deficiency corresponds to levels below 180 pg/ml

Individuals with CEH had significantly lower NRS scores than those with migraine (7.04 ± 1.37 vs. 8.54 ± 1.43 , $p < 0.001$), as well as significantly more monthly headache days (17.07 ± 9.53 vs. 13.33 ± 9.38 , $p = 0.003$) and significantly less acute medication days (5.45 ± 5.13 vs. 11.42 ± 8.90 , $p < 0.001$), and significantly less often presented with medication overuse headache (2.68% vs. 26.61%, $p < 0.001$).

Patients with CEH were diagnosed with vitamin B12 deficiency significantly more often than individuals with migraine (56.86% vs. 9.38%, $p < 0.001$) and had significantly lower vitamin B12 levels than the latter (237.39 ± 167.52 vs. 432.54 ± 229.46 pg/ml, $p < 0.001$).

CEH was associated with a family history of headache significantly less often than migraine (33.04% vs. 75.89%, $p < 0.001$).

The proportion of smokers among individuals with CEH was significantly higher than in the migraine group (43.75% vs. 10.71%, $p < 0.001$). While patients with CEH declared being regular coffee drinkers significantly less often than individuals with migraine (65.18% vs. 77.68%, $p = 0.038$), they dominated over the latter group in terms of the mean number of coffee cups per day (2.20 ± 1.08 vs. 1.82 ± 0.93 cups, $p = 0.015$).

The percentage of belly sleepers in the CEH group was significantly higher than in the migraine group (40.18% vs. 14.29%, $p < 0.001$). While the proportion of side sleepers was significantly lower in the CEH group than in the migraine group (35.71% vs. 60.71%, $p < 0.001$), side sleepers from the CEH group slept exclusively on one side significantly more often than their migraine counterparts (72.50% vs. 30.88%, $p < 0.001$). Similarly, the CEH group included a significantly larger proportion of individuals who slept either exclusively on one side or in a prone position with their head twisted exclusively to one side (74.12% vs. 35.71%, $p < 0.001$).

All relationships described above were confirmed as statistically significant in a series of univariate logistic regression analyses (Table 3).

A multivariate logistic regression analysis was conducted to identify variables that could be considered in the prevention of CEH. The regression model included three parameters that distinguished patients with CEH from individuals with migraine at $p \leq 0.05$ and were potentially modifiable, i.e., belly sleeping, smoking, and vitamin B12 deficiency. The model explained over 42% of variance in the occurrence of CEH (McFadden's

Table 4 Independent determinants of CEH: results of multivariate logistic regression analysis

Variable	OR	95%CI	p-value
Active smoking	22.61	2.54–200.98	0.005
Belly sleepers	11.10	2.02–60.86	0.006
Vitamin B12 deficiency	8.58	1.93–38.10	0.005

McFadden's pseudo-R²=0.4264. Uncentered VIF values for explanatory variables: active smoking 1.56, belly sleepers 1.53, vitamin B12 deficiency 1.68. Vitamin B12 deficiency corresponds to levels below 180 pg/ml

pseudo-R² = 0.4264). All three explanatory variables were identified as independent determinants of CEH. No multicollinearity was found between the explanatory variables, as shown by uncentered VIF values between 1.53 and 1.68 (Table 4).

Discussion

The present study, based on an extensive comparison of age- and sex-matched groups, demonstrated that individuals with CEH differed significantly from those with migraine in terms of various clinical, demographic, and lifestyle characteristics. These findings contribute importantly to the limited body of knowledge on CEH. Our study supports evidence that CEH is a separate clinical entity. Moreover, some of the variables identified herein could be potentially considered for the prevention of this condition.

Our analysis demonstrated that CEH was diagnosed at a significantly older age than migraine. This finding is not surprising given that CEH is a secondary condition, whereas migraine is a disorder with a significant genetic component [17]. Indeed, in the present study, the proportion of patients with CEH who had a family history of headache was significantly smaller than in the migraine group. Interestingly, however, patients with CEH had significantly shorter duration of disease as compared with individuals with migraine. The mean duration of disease was merely 4.08 years for the CEH group compared with 17.74 years in the migraine group, with some individuals with CEH presenting as early as one month after the onset of pain. Thus, early presentation to a specialist might be considered the first prompt that a person with a headache suffers from CEH rather than migraine.

The present study identified several clinical characteristics that significantly differed CEH and migraine. While most individuals with CEH presented with side-locked headache, one in eight of them reported bilateral pain. This implies that the criterion of “preferential unilaterality” [9] is not entirely reliable. Even if we assume that CEH derives from an injury to the cervical region, some individuals may present with bilateral trauma and hence, suffer from bilateral pain. Indeed, while generally agreed upon, the “preferential unilaterality” criterion no longer exists in the current version of the ICHD-3 [7].

Further, most patients with CEH reported neck pain occurring independently of headache (76.92%), and a considerable proportion of this group presented with GON sensitivity to palpation (85.44%). These findings imply that the headache observed in these patients with CEH indeed originated from the cervical region. Although this observation may seem trivial and redundant for some, it should be emphasized that due to a convergence of sensory information in the trigeminocervical nucleus, the sensation of neck pain present in some individuals with migraine can be not infrequently misdiagnosed as CEH, whereas the latter can be misinterpreted as migraine [12–14]. Indeed, in the present study, neck pain independent of headache and GON sensitivity to palpation were also found in 39.56% and 30.21% of individuals with migraine, respectively.

Based on the analysis of NRS scores, pain associated with CEH appeared to be less burdensome than in migraine, despite a significantly higher number of monthly headache days. The relatively lower intensity of CEH could probably explain the lower consumption of acute medications by patients with this condition, as shown by a significantly lower number of acute medication days than in the migraine group. Alternatively, available acute medications for CEH were not as effective as expected, and hence, were taken less often. Our findings also imply that patients with CEH have a lower risk of medication overuse headache, either due to lower consumption of acute medications or due to another yet unidentified mechanism specific to this type of headache.

Importantly, this study identified several lifestyle factors distinguishing patients with CEH from individuals with migraine. A substantial proportion of patients with CEH reported sleeping exclusively in the prone position. The so-called belly sleepers sleep lying with their head twisted to either side, which, with no doubt, has a fundamental effect on neck biomechanics, especially if repeated every night for years. Sleeping in a prone position requires the neck to be twisted 90 degrees, which results in extreme biomechanical stress [18]. Specifically, belly sleeping causes increased spinal disc pressure, extra compression of the cervical zygapophysial joints, muscle imbalance, and reduced muscle perfusion [19]. The sternocleidomastoid muscle becomes overstretched on one side, while the contralateral trapezius muscle is compressed. Muscle imbalance also involves the underlying suboccipital muscles, leading to their stiffness, reduced range of motion, and resultant narrowing of the suboccipital triangle [20]. Reduced blood perfusion is an additional factor contributing to both increased muscle tension within the cervical region and cervical nerve injury [21, 22]. Nociceptive input from the cervical structures is transmitted via afferent nerve fibers of the trigeminocervical nucleus: C1–C3 spinal nerves, GON,

the third occipital nerve, and the C2 dorsal root ganglion [1]. Additionally, these nerves can be compressed by the stiffened muscles or even entrapped, as some imaging studies demonstrated cervical nerve edema ipsilaterally to unilateral occipital neuralgia [21, 22]. Within the trigeminocervical nucleus, sensory cervical nerves converge with sensory fibers of the trigeminal nerve, through which pain signals from cervical structures are delivered to the brain and misinterpreted as pain originating from the head [1]. Given the anatomical considerations discussed above, belly sleeping appears as a potential potent trigger of CEH. However, to the best of our knowledge, aside from some non-evidence-based recommendations published online, the role of belly sleeping has not been implicated in the pathogenesis of CEH thus far. This is quite surprising considering that belly sleeping has already been identified as an established contributor to pain in other anatomical regions, such as the shoulders and lower back [23, 24].

Our analysis demonstrated that side sleeping was significantly less common in individuals with CEH than in those with migraine. This observation may be surprising as, similar to belly sleeping, side sleeping also causes asymmetry in neck biomechanics. However, biomechanical disturbances associated with side sleeping are less severe, with neck strain, if any, originating primarily from the scalene muscles and levator scapulae [20]. Aside from C3, these muscles receive innervation from lower segments of the spine, and hence, sensory information delivered from these structures to the trigeminocervical nucleus is limited. Thus, disturbances associated with side sleeping may not necessarily be a sufficient trigger for CEH. Our analysis sheds some new light on these findings. While side sleeping was significantly less common among individuals with CEH than in those with migraine, the CEH group contained a significantly larger proportion of persons sleeping exclusively on one side. Repetitive unilateral loading may therefore be sufficient to induce persistent muscle stiffness and contribute to CEH, whereas changing sleeping posture from one side to another overnight may have a protective effect.

In our study, patients with CEH were diagnosed with vitamin B12 deficiency significantly more often than individuals with migraine and presented with significantly lower levels of this vitamin. According to the literature, vitamin B12 deficiency occurs in 2–3% of adults in the United States [25], being the most common dietary deficiency observed in this population [26]. Vitamin B12 plays a crucial role in neurological health, and its pivotal role has already been recognized in other neurological and psychiatric conditions or symptoms, including sciatica, peripheral neuropathy, ataxia, brain fog, depression, cognitive dysfunction, and dementia [25, 27]. In regards to our findings, the role that vitamin B12 plays in

nerve myelination is particularly worth emphasizing [28]. Because of impaired remyelination of cervical nerves and resultant delayed recovery, vitamin B12-deficient patients with CEH might be more prone to nerve damage during repeated injury associated with sleeping in a prone position. However, to the best of our knowledge, none of the previously published studies dealt with the role of vitamin B12 in CEH pathogenesis. The risk of vitamin B12 deficiency increases in older adults, primarily due to malabsorption [29]. The fact that in our present study, CEH was developed at a significantly older age than migraine implies that an age-related vitamin B12 deficiency might have been involved in some cases.

Additionally, the CEH group included a significantly larger proportion of active smokers than the migraine group. Smoking is an established risk factor for vitamin B12 deficiency [30]. However, our multivariate logistic regression analysis identified both smoking and vitamin B12 deficiency as independent determinants of CEH. Thus, smoking might contribute to CEH through other mechanisms. Indeed, high doses of nicotine were already long ago shown to acutely affect sensory and neuromuscular transmission via nicotine receptors located on axons and terminals of many sensory and motor nerve fibers [31]. Although a causal relationship between smoking and CEH has not been established yet, one cross-sectional study identified smoking as an independent predictor of persistent headache following surgical treatment for CEH [32]. However, given the cross-sectional design of both our study and the study mentioned above, causality should not be inferred.

The results presented herein, in particular, the outcome of a multivariate logistic regression analysis, might have some implications for the prevention and management of CEH. All three variables included in our model, belly sleeping, smoking, and vitamin B12 deficiency, turned out to be independent determinants of CEH. This implies that all individuals with this condition should be advised to avoid sleeping in the prone position, and optimally, also unilateral side sleeping. Further, all individuals with CEH should be screened for vitamin B12 deficiency and receive appropriate supplementation whenever needed. A recent meta-analysis demonstrated that an intervention in the form of vitamin B12 supplementation might produce a beneficial effect on neurological function [27]. However, optimal dosing and duration of the supplementation are yet to be established. Finally, although the direction of causal relationship between smoking and CEH is yet to be determined, smoking cessation should also be recommended as a part of the CEH management plan.

This study has some potential limitations. First, due to its cross-sectional design, we could not determine the causal direction of the identified relationships. Yet the

direction was obvious in most cases and supported by published evidence. Second, while most individuals with migraine used e-diaries, some headache characteristics in CEH patients were retrospectively self-reported, and, therefore, prone to reporting bias. Third, the prior history of neck/head trauma was available solely for CEH patients and thus, not suitable for comparative analysis. Fourth, some potentially important variables, such as disability, cervical musculoskeletal impairments, signs and symptoms of sensitization, and psychosocial factors, were not collected. Fifth, the term “dominant” sleeping position might have been vague, and some participants might not have reported their sleeping posture accurately. Finally, we did not ask our participants if they used generic pillows, pillows dedicated to a given sleeping posture, or if they did not use pillows at all, which might influence sleep posture and cervical biomechanics.

Conclusions

CEH differs from migraine in several clinical, demographic, and lifestyle characteristics. Presence of neck pain independent of headache, GON sensitivity to palpation, older age at headache onset, and shorter interval between headache onset and referral might be diagnostic prompts in suspected CEH. Belly sleeping, vitamin B12 deficiency, and smoking appear to be independent determinants of CEH and might be considered in the prevention and management of this condition.

Abbreviations

ICHD-3	3rd edition of the IHS International Classification of Headache Disorders
95%CI	95% Confidence interval
CEH	Cervicogenic headache
GON	Greater occipital nerve
MRI	Magnetic resonance imaging
NRS	Numeric rating scale
OR	Odds ratio
SD	Standard deviation
VIF	Variance inflation factor

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Author contributions

Conceptualization and study design: M.N. Data acquisition and data management: M.N., M.S., S.K.-S., M.W.-P., and A.G.-D. Statistical analyses: M.N. Writing – original draft preparation: M.N. and A.G.-D. Writing – review & editing: All authors. Visualization: M.N. Project administration: M.N.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study protocol was approved by the Local Ethics Committee at the Ludwik Rydygier Collegium Medicum in Bydgoszcz. No written consent

was required due to the retrospective character of the study. The study was conducted in accordance with the Declaration of Helsinki.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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